a loss of ₹ 8,000 with a probability of $0 \cdot 38$. Find the expected gain.

7. What are the properties of Mathematical expectation?
8. What do you understand by Mathematical expectation?
9. How do you define variance in terms of Mathematical expectation?
10. Define Mathematical expectation in terms of discrete random variable.
11. State the definition of Mathematical expectation using continuous random variable.
12. In a business venture a man can make a profit of ₹ 2,000 with a probability of 0.4 or have a loss of ₹ 1,000 with a probability of $0 \cdot 6$. What is his expected, variance and standard deviation of profit?
13. The number of miles an automobile tire lasts before it reaches a critical point in tread wear can be represented by a p.d.f.

$$f(x)=\begin{cases}\frac{1}{30}e^{-\frac{x}{30}}, & for\ x>0\\ 0 \quad\quad , & for\ x\le 0\end{cases}$$

Find the expected number of miles (in thousands) a tire would last until it reaches the critical tread wear point.

14. A person tosses a coin and is to receive ₹ 4 for a head and is to pay ₹ 2 for a tail. Find the expectation and variance of his gains.
15. Let X be a random variable and $Y = 2X + 1$. What is the variance of Y if variance of X is 5 ?

Exercise 6.3

Choose the correct Answer

1. Value which is obtained by multiplying possible values of random variable with probability of occurrence and is equal to weighted average is called
   (a) discrete value
   (b) weighted value
   (c) expected value
   (d) cumulative value

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2. Demand of products per day for three days are 21, 19, 22 units and their respective probabilities are $0 \cdot 29, 0 \cdot 40, 0 \cdot 35$. Pofit per unit is $0 \cdot 50$ paisa then expected profits for three days are
   (a) 21, 19, 22 (b) 21.5, 19.5, 22.5
   (c) 0.29, 0.40, 0.35 (d) 3.045, 3.8, 3.85
3. Probability which explains x is equal to or less than particular value is classified as
   (a) discrete probability
   (b) cumulative probability
   (c) marginal probability
   (d) continuous probability
4. Given $E(X) = 5$ and $E(Y) = -2$, then $E(X - Y)$ is
   (a) 3 (b) 5 (c) 7 (d) –2
5. A variable that can assume any possible value between two points is called
   (a) discrete random variable
   (b) continuous random variable
   (c) discrete sample space
   (d) random variable
6. A formula or equation used to represent the probability distribution of a continuous random variable is called
   (a) probability distribution
   (b) distribution function
   (c) probability density function
   (d) mathematical expectation
7. If X is a discrete random variable and $p(x)$ is the probability of X, then the expected value of this random variable is equal to

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